FILTER DESIGN TEST CASE

Duplexer Module - Band 25 (1900 MHz) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting Band 25 (1900 MHz) for Millimeter-Wave Beamforming. The filter is designed on AlN on SiC substrate with a target center frequency of 2144.6 MHz and bandwidth of 156.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: Band 25 (1900 MHz)
- Application: Millimeter-Wave Beamforming
- Substrate: AlN on SiC
- Package: Flip-Chip BGA
- Design Tool: PathWave EM Design
- Design Center: Irvine Design Center

This specification applies to all variants of the Duplexer Module design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 2144.6 MHz
Bandwidth (3 dB): 156.4 MHz
Insertion Loss Target: <= 3.4 dB
Return Loss Target: >= 13.0 dB
Out-of-Band Rejection: >= 44.0 dB
Isolation (Duplexer): >= 41.0 dB
Q Factor: 7073
Temperature Coefficient: -23.8 ppm/°C
Die Size: 1.7 x 2.3 mm
Wafer Size: 6-inch
Number of Resonators: 10
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Duplexer Module template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 2144.6 MHz, BW 156.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on SiC).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6434 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.4 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 235 MHz offset >= 44.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 6.4 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-03-25
Devices Tested: 25
Insertion Loss (meas): 3.08 dB
Return Loss (meas): 14.6 dB
Rejection (meas): 46.6 dB
Isolation (meas): 46.2 dB
Q Factor (meas): 7073
Group Delay Variation: 6.4 ns
Power Handling: +36 dBm
Die Yield: 94.0%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The Duplexer Module design demonstrated below-target performance for Band 25 (1900 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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